Software Requirements Specification

for

Real-Time Multi-Vendor Food Ordering and Delivery Management System for a University Campus

Version 1.0

Prepared by

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Revisions

Version	Primary Author(s)	Description of Version	Date Completed
0.1	Ritik Ahirwar , Pravin Tanpure	Initial draft of the Software Requirements Specification introduction, including product scope, intended audience, definitions, document conventions, and references	18/01/26
0.2	Vikas Saini , Abhimanyu Singh Rawat	Added the Overall Description section, including product overview, system functionality, design and implementation constraints, and assumptions and dependencies for the campus food ordering and delivery system	20/01/26
0.3	Banoth Venkatamaniram, Kajal	Added detailed external interface requirements, functional requirements, and use case models defining user, vendor, and delivery personnel interactions with the system.	21/01/26
0.4	Ashish Kumar Soni, Banoth Ramcharan	Added non-functional requirements, including performance, security, safety, and software quality attributes such as reliability, usability, maintainability, scalability, and portability	22/01/26
1.0	Aarsh Kaushik	Finalized the Software Requirements Specification by incorporating all remaining system constraints and requirements, including database, legal and policy, internationalization, reusability, and backup and recovery requirements. Added a complete data dictionary as an appendix to formally define core system data elements. This version represents the finalized SRS baseline for evaluation and implementation	22/01/26

1 Introduction

1.1 Product Scope

The proposed software is a **campus-wide food ordering and delivery platform** designed to unify multiple on-campus food outlets into a single, easy-to-use digital ecosystem. Its primary purpose is to allow campus residents—such as students and staff—to conveniently browse menus, place orders, and track deliveries in real time from a single application. By integrating food vendors and delivery agents into the same system, the platform eliminates the need for multiple standalone ordering methods, reduces manual coordination, and improves overall efficiency of campus food services.

The product aims to enhance user convenience, transparency, and speed while optimizing operations for vendors and delivery personnel. Key benefits include real-time order and location tracking for users, streamlined menu and order management for vendors, and clear task assignment with GPS-based navigation for delivery agents. The overall goal of the system is to create a reliable, scalable, and responsive food delivery infrastructure tailored to a campus environment, improving user satisfaction, reducing order errors, and enabling data-driven management of campus food operations.

1.2 Intended Audience and Document Overview

This SRS document is intended for developers, project managers, testers, end users, and documentation writers involved in the campus food ordering and delivery system. Developers and testers will use it to understand and verify functional and non-functional requirements, while project managers will use it to plan, monitor, and manage the project. End users and documentation writers can refer to the document to understand system features and expected behaviour.

The document is organized from high-level concepts to detailed requirements. It begins with an introduction and overall system description, followed by functional requirements for users, vendors, and delivery personnel, and then non-functional and interface requirements. Readers are advised to start with the overview sections, after which developers and testers should focus on the detailed requirements, while non-technical stakeholders can primarily refer to the high-level sections.

1.3 Definitions, Acronyms and Abbreviations

The following abbreviations and acronyms are used throughout this Software Requirements Specification (SRS). They are listed in alphabetical order.

Abbreviation	Definition
API	Application Programming Interface
GPS	Global Positioning System
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
SRS	Software Requirements Specification
UI	User Interface
URL	Uniform Resource Locator
WebSocket	A communication protocol that provides full-duplex, real-time communication between client and server

1.4 Document Conventions

This document follows the **IEEE Software Requirements Specification (SRS) standard** formatting guidelines. The document is written using **Arial font**, with a font size of **11 or 12** for body text. All text is **single-spaced** and maintains **1-inch margins** on all sides, as defined by the provided template. Section and subsection numbering and titles strictly follow the template structure.

The following conventions are used throughout the document:

- **Bold text** is used for section and subsection headings.
- *Italic text* is used for comments, clarifications, or notes.
- Numbered lists indicate ordered steps or priorities.
- Bullet points are used for unordered lists of features or requirements.
- All requirements are written in clear, declarative language to avoid ambiguity.

1.5 References and Acknowledgments

The following documents and resources were referenced during the preparation of this SRS:

1. IEEE Std 830-1998 / IEEE Std 29148-2018 – IEEE Recommended Practice for Software Requirements Specifications
2. Course material and lecture notes on Software Development and Operations

3. WebSocket Protocol Specification (RFC 6455)

4. General documentation on RESTful APIs and real-time system design

Acknowledgment is given to course instructors and teaching assistants for providing guidance on SRS structure and standards.

2 Overall Description

2.1 Product Overview

The campus food ordering and delivery system described in this SRS is a **new, self-contained software product** designed to address inefficiencies in existing campus food services. Traditionally, campus residents rely on physical queues, phone-based ordering, or vendor-specific applications, which results in fragmented user experiences, limited order visibility, and poor coordination between vendors and delivery personnel. This product is introduced as a unified digital platform that consolidates multiple campus food outlets, delivery agents, and users into a single, integrated system.

The system operates as a **centralized backend platform** with multiple client-facing applications. It includes a user application for students, a vendor dashboard for food outlets, and a delivery module for delivery personnel. These components interact through backend services responsible for authentication, order management, vendor management, and real-time communication. The system integrates real-time technologies such as WebSockets and GPS-based location tracking to ensure instant updates of order status and delivery location. While the product is standalone, it can interface with external services such as map providers for location visualization and notification services for alerts, making it extensible and scalable within a broader campus digital ecosystem.

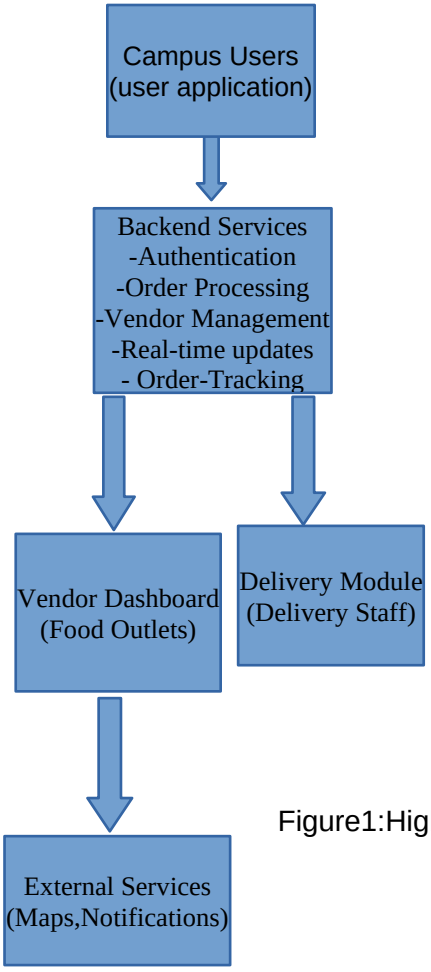

Figure1:High Level System-Context Diagram

2.2 Product Functionality

The major functions of the campus food ordering and delivery system include:

- Allow users to register, authenticate, and manage their accounts
- Enable users to browse menus from multiple campus food vendors
- Allow users to place food orders and make order requests
- Provide real-time updates on order status (e.g., accepted, preparing, out for delivery)
- Display real-time delivery location tracking on a map
- Allow vendors to register, manage their profiles, and update menus and prices
- Enable vendors to receive, accept, and process incoming orders
- Allow delivery personnel to receive assigned delivery tasks
- Transmit and track live GPS location data from delivery personnel
- Facilitate real-time communication between system components
- Maintain centralized order, user, vendor, and delivery data

2.3 Design and Implementation Constraints

The campus food ordering and delivery system is subject to the following constraints:

- The system must support web and mobile clients commonly used within the campus environment.
- The backend must run on standard server hardware with limited memory and processing resources.
- Real-time functionality is constrained to the use of WebSockets or similar event-driven communication mechanisms.
- A centralized database must be used to ensure consistency of user, vendor, and order data.
- The system may integrate with third-party services such as map and notification providers, whose availability may affect system behavior.

- All communication must use standard and secure Internet protocols such as HTTP/HTTPS.
- Basic security practices, including authentication and encrypted communication, must be enforced.
- The system must support concurrent usage by multiple users, vendors, and delivery personnel.
- The software must follow standard coding and documentation conventions to allow future maintenance.

2.4 Assumptions and Dependencies

The following assumptions and dependencies may significantly affect the design and operation of the system:

- It is assumed that the campus administration approves the introduction of a dedicated food delivery mechanism, even if no such delivery system currently exists on campus.
- It is assumed that food vendors operating on campus are willing to register on and actively use the platform.
- It is assumed that users have access to smartphones or web-enabled devices with a stable Internet connection.
- The system depends on the availability and reliability of third-party services such as map and location providers for real-time tracking.
- The system assumes access to a hosting environment capable of supporting real-time communication and concurrent users.

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

The User Application interface allows campus users to browse food vendors, view menus, place orders, and track deliveries. Interaction is primarily menu-driven, using lists, buttons, and a map view for real-time tracking. Users can view order status updates such as accepted, preparing, and out for delivery through a dedicated order tracking screen.

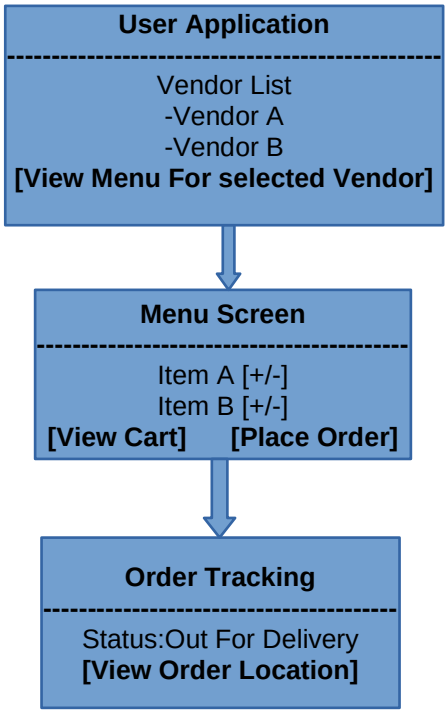

Figure 2 : User Interface

The Vendor Dashboard interface enables food vendors to manage their menus and process incoming orders. Vendors can view new orders, update order status, and modify menu items and prices. The interface is primarily table-based, allowing quick status updates through simple controls.

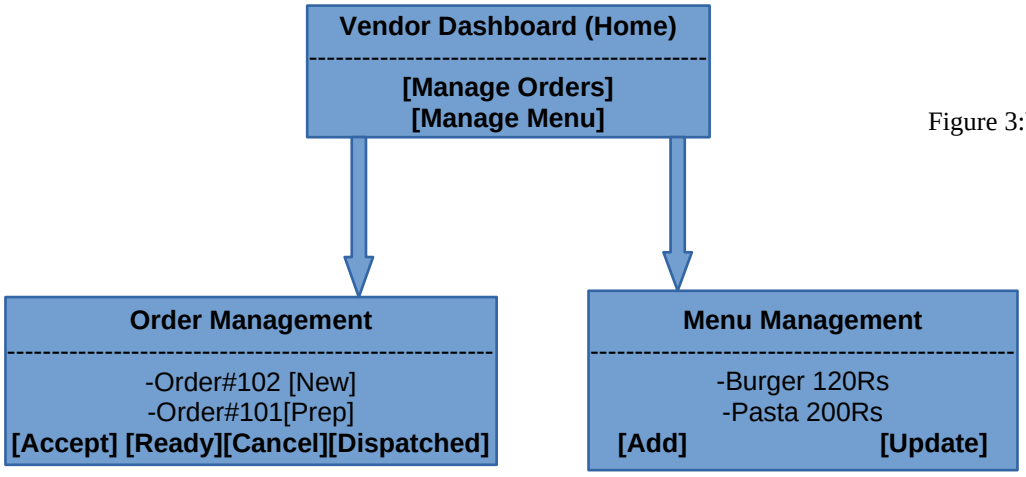

Figure 3:Vendor Dashboard

The Delivery Module interface allows delivery personnel to receive assigned orders and navigate to pickup and delivery locations. The interface focuses on minimal interaction and real-time location tracking. GPS data is transmitted automatically, while delivery personnel update order completion status through simple actions.

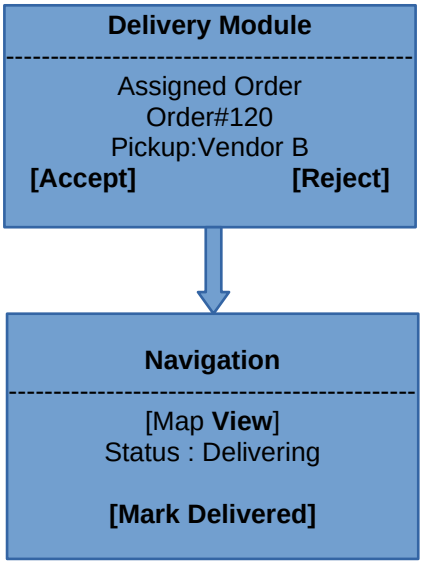

Figure 4: Delivery Module

3.1.2 Hardware Interfaces

The campus food ordering and delivery system interacts with standard hardware components commonly available to users, vendors, and delivery personnel. No specialized or custom hardware is required.

- **User Devices:** Smartphones, tablets, and personal computers used by campus users to access the user application. These devices provide input through touch or keyboard and display order information and tracking details.
- **Vendor Devices:** Desktop computers, laptops, or tablets used by food vendors to access the vendor dashboard for order and menu management.
- **Delivery Devices:** GPS-enabled smartphones used by delivery personnel to receive delivery assignments, provide location updates, and navigate to pickup and delivery locations.
- **Location Hardware (GPS):** Built-in GPS sensors on delivery devices are used to capture and transmit real-time location data for delivery tracking.
- **Server Hardware:** Backend services are hosted on standard server hardware or cloud infrastructure responsible for processing requests, storing data, and handling real-time communication.

3.1.3 Software Interfaces

The campus food ordering and delivery system interacts with several external and internal software components to provide its functionality. These interfaces support data exchange, real-time updates, and integration with supporting services.

- **Authentication Service:** The system interfaces with an authentication component to manage user, vendor, and delivery personnel login and authorization.
- **Database Management System:** The backend interfaces with a centralized database to store and retrieve user profiles, vendor information, menus, orders, and delivery status data.
- **Mapping and Location Services:** The system interfaces with third-party mapping software to display maps and visualize real-time delivery locations for users and delivery personnel.
- **Notification Services:** External notification services are used to send alerts and updates to users and vendors regarding order status changes.
- **Real-Time Communication Services:** The system interfaces with real-time messaging components (e.g., WebSocket-based services) to propagate order status updates and live location data to connected clients.
- **Web and Mobile Client Applications:** The backend interfaces with web and mobile client software used by users, vendors, and delivery personnel to exchange requests and responses.

3.2 Functional Requirements

3.2.1 F1: User Authentication

The system shall allow users, vendors, and delivery personnel to authenticate themselves before accessing system functionalities.

3.2.2 F2: Vendor Browsing

The system shall allow authenticated users to view a list of registered campus food vendors.

3.2.3 F3: Menu Viewing

The system shall allow users to view the menu, prices, and availability of items offered by a selected vendor.

3.2.4 F4: Order Placement

The system shall allow users to select food items and place an order with a chosen vendor.

3.2.5 F5: Order Status Tracking

The system shall allow users to view order status updates, including accepted, preparing, dispatched, and delivered states.

3.2.6 F6: Delivery Location Tracking

The system shall allow users to view the real-time location of the delivery personnel after the order is dispatched.

3.2.7 F7: Vendor Order Management

The system shall allow vendors to view incoming orders and update order statuses.

3.2.8 F8: Vendor Order Cancellation

The system shall allow vendors to cancel an order if it cannot be fulfilled and notify the user of the cancellation.

3.2.9 F9: Vendor Menu Management

The system shall allow vendors to add and update menu items and prices.

3.2.10 F10: Delivery Assignment

The system shall assign orders ready for delivery to available delivery personnel.

3.2.11 F11: Delivery Acceptance

The system shall allow delivery personnel to accept or reject assigned delivery requests.

3.2.12 F12: Delivery Status Update

The system shall allow delivery personnel to update delivery status, including marking an order as delivered.

3.2.13 F13: Real-Time Notifications

The system shall provide real-time notifications to users, vendors, and delivery personnel when order or delivery status changes occur.

3.3 Use Case Model

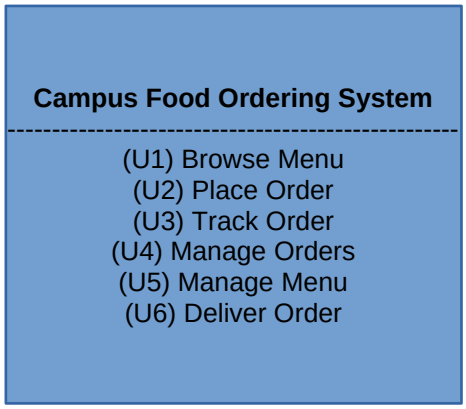

Figure 5: Use cases Diagram

3.3.1 Use Case U1-U3: Place Order

Author: Aarsh Kaushik, Banoth Ramcharan , Ashish Kumar Soni

Purpose:
To allow a campus user to select food items from a vendor and place an order.

Requirements Traceability:
F2, F3, F4

Priority:
High

Preconditions:

- User is authenticated
- Vendor is registered and available

Postconditions:

- Order is created in the system
- Vendor is notified of the new order

Actors:

- User

Exceptions:

- Selected items become unavailable
- Order submission fails due to network issues

Includes:
None

Notes/Issues:
Payment handling is assumed to be handled externally or out of scope.

3.3.2 Use Case U4-U5: Manage Orders (Vendor)

Author: Ritik Ahirwar , Pravin Tanpure , Vikas Saini

Purpose:
To allow vendors to view incoming orders and update their status.

Requirements Traceability:
F7, F8

Priority:
High

Preconditions:

- Vendor is authenticated
- Orders exist in the system

Postconditions:

- Order status is updated or order is canceled
- User is notified of the change

Actors:

- Vendor

Exceptions:

- Order cannot be fulfilled
- System fails to update order state

Includes:
None

Notes/Issues:
Canceled orders may require reassignment or notification handling.

3.3.3 Use Case U6: Deliver Order

Author: Kajal, Banoth Venkatamaniram, Abhimanyu Singh Rawat

Purpose:
To allow delivery personnel to deliver an assigned order and update its status.

Requirements Traceability:
F10, F11, F12

Priority:
Medium

Preconditions:

- Order is marked ready for delivery
- Delivery personnel is available and authenticated

Postconditions:

- Order is marked as delivered
- User is notified of completion

Actors:

- Delivery Personnel

Exceptions:

- Delivery request is rejected
- Delivery fails due to unforeseen issues

Includes:
None

Notes/Issues:
Rejected deliveries are reassigned to other delivery personnel.

4 Other Non-functional Requirements

4.1 Performance Requirements

The campus food ordering and delivery system must meet the following performance requirements to ensure a responsive and reliable user experience under normal campus operating conditions.

- **Response Time:**
  The system shall respond to user interactions such as menu browsing, order placement, and status queries within **2 seconds** under normal load.
- **Order Processing Time:**
  The system shall record and acknowledge a placed order within **3 seconds** and make it visible to the vendor dashboard.
- **Real-Time Status Updates:**
  Changes in order status initiated by vendors or delivery personnel shall be propagated to all relevant users within **1 second**.
- **Location Tracking Update Rate:**
  The delivery personnel's location shall be updated and reflected on the user interface at least once every **5 seconds** during active delivery.
- **Concurrent Users:**
  The system shall support **at least 200 concurrent users**, including users, vendors, and delivery personnel, without significant degradation in performance.
- **System Availability:**
  The system shall maintain an operational availability of **99%** during peak campus hours.

4.2 Safety and Security Requirements

The campus food ordering and delivery system shall incorporate safety and security measures to prevent misuse, protect user data, and minimize potential harm resulting from system failures or unauthorized access.

Security Requirements

- The system shall require all users, vendors, and delivery personnel to authenticate themselves before accessing any system functionality.

- The system shall ensure that authenticated users can only access features permitted by their assigned role (user, vendor, or delivery personnel).
- The system shall protect sensitive data such as user credentials and personal information from unauthorized access.
- The system shall ensure secure communication between clients and backend services to prevent data interception or tampering.
- The system shall log critical actions such as order placement, cancellation, and delivery status updates for auditing and troubleshooting purposes.

Privacy Requirements

- The system shall limit access to user personal information to only those system components and actors that require it for order fulfillment.
- The system shall ensure that delivery location data is visible only during active deliveries and only to authorized users.

Safety Requirements

- The system shall prevent unauthorized modification or deletion of active orders.
- The system shall notify users and vendors in real time if an order is cancelled or fails due to system or operational issues.
- The system shall handle system failures gracefully by preserving existing order data and preventing inconsistent order states.

Authentication and Access Control

- The system shall enforce unique user identities for all users, vendors, and delivery personnel.
- The system shall prevent unauthenticated users from accessing protected system resources.
- The system shall terminate or restrict access if suspicious or unauthorized activity is detected.

Regulatory and Policy Considerations

- The system shall comply with general institutional data protection and privacy policies applicable within the campus environment.
- No external safety certifications are required for the initial deployment of the system.

4.3 Software Quality Attributes

4.3.1 Reliability

The system shall operate reliably during normal campus operating hours to ensure uninterrupted food ordering and delivery services.

- The system shall maintain an operational availability of at least **99%** during peak usage periods.
- The system shall preserve order and delivery data in the event of partial system failures to prevent data loss or inconsistent order states.
- The system shall recover from transient failures without requiring user intervention wherever possible.

These requirements ensure that users and vendors can depend on the system for critical ordering operations.

4.3.2 Usability

The system shall prioritize ease of use over ease of learning to ensure efficient interaction for all user roles.

- The system shall provide a menu-driven user interface with clear navigation and minimal steps to complete common tasks such as placing or accepting orders.
- The system shall allow users to complete core actions (e.g., placing an order or updating order status) with no more than **three primary interactions**.
- The system shall present real-time feedback for user actions such as order placement, cancellation, and delivery updates.

These measures ensure that users, vendors, and delivery personnel can interact with the system quickly and intuitively.

4.3.3 Maintainability

The system shall be designed to allow future maintenance and enhancement with minimal effort.

- The system shall use a modular backend architecture, where components such as authentication, order management, and location tracking are logically separated.
- The system shall follow consistent coding standards and include documentation for major components to support future developers.

- The system shall allow individual components to be updated or replaced without requiring changes to the entire system.

This approach ensures long-term sustainability of the software beyond initial deployment.

4.3.4 Scalability

The system shall be capable of scaling to support increased usage as campus demand grows.

- The system shall support a minimum of **200 concurrent users** without significant degradation in response time.
- The system shall allow horizontal scaling of backend services to handle increased order volume and real-time updates.

These requirements ensure that the system remains functional as the number of users, vendors, or delivery personnel increases.

4.3.5 Portability

The system shall be portable across commonly used platforms within the campus environment.

- The user and vendor interfaces shall be accessible through standard web browsers without requiring specialized hardware or software.
- The delivery module shall operate on standard GPS-enabled smartphones.

This ensures broad accessibility and ease of deployment.

5 Other Requirements

Database Requirements: *The system shall use a centralized database to store user profiles, vendor information, menus, orders, and delivery data. The database shall ensure data consistency and integrity for concurrent read and write operations and shall retain order and delivery records for audit and troubleshooting purposes.*

Internationalization Requirements: *The system shall support display of prices and basic text in formats suitable for the campus locale and shall be designed to allow future extension to support multiple languages without major architectural changes.*

Legal and Policy Requirements: *The system shall comply with applicable campus policies regarding data privacy and acceptable use of digital services. User data shall be collected and used only for purposes related to food ordering and delivery.*

Reusability Requirements: *The system shall be designed using reusable software components so that modules such as authentication, order management, and notification services can be reused in future campus applications.*

Backup and Recovery Requirements: *The system shall perform periodic backups of critical data to prevent loss in case of system failure and shall allow restoration of essential data to a consistent state after recovery.*

Appendix A – Data Dictionary

1.1 Data Dictionary

The following table describes the key constants, state variables, inputs, and outputs used in the campus food ordering and delivery system. It includes their descriptions, possible values, and related operations or requirements.

Item Name	Type	Description	Possible Values / Format	Related Operations / Requirements
UserID	Input	Unique identifier for a user	Integer / UUID	Used in authentication, order placement (F1, F4)
VendorID	Input	Unique identifier for a vendor	Integer / UUID	Used in menu viewing, order management (F3, F7)
DeliveryAgentID	Input	Unique identifier for delivery personnel	Integer / UUID	Used in delivery assignment and tracking (F10, F11)
OrderID	State Variable	Unique identifier for an order	Integer / UUID	Used across order lifecycle (F4–F12)
OrderStatus	State Variable	Current status of an order	New, Accepted, Preparing, Ready, Dispatched, Delivered, Cancelled	Updated by vendor and delivery personnel (F5, F7, F8, F12)
MenuItem	State Variable	Food item offered by a vendor	Name, Price, Availability	Managed by vendor (F3, F9)
CartItems	State Variable	Items selected by user before ordering	List of MenuItems	Used during order placement (F4)
OrderTimestamp	State Variable	Time at which order was placed	DateTime	Used for tracking and audit
DeliveryLocation	State Variable	Real-time location of delivery personnel	Latitude, Longitude	Used for live tracking (F6, F12)
AuthenticationStatus	State Variable	Authentication state of a user	Authenticated, Unauthenticated	Controls access to system features (F1)
AcceptOrder	Input	Vendor action to accept an order	Boolean	Changes OrderStatus to Accepted (F7)
CancelOrder	Input	Vendor action to cancel an order	Boolean	Changes OrderStatus to Cancelled (F8)
AssignDelivery	Output	Assignment of order to delivery personnel	OrderID, DeliveryAgentID	Triggered when order is ready (F10)
DeliveryDecision	Input	Delivery personnel response to	Accepted, Rejected	Affects delivery workflow (F11)

Item Name	Type	Description	Possible Values / Format	Related Operations / Requirements
		assignment		
MarkDelivered	Input	Delivery personnel action confirming delivery	Boolean	Changes OrderStatus to Delivered (F12)
Notification	Output	System-generated alert	Text message / Push notification	Sent on status changes (F13)
Price	Constant	Cost of a menu item	Currency value	Displayed during menu viewing (F3)
MaxConcurrentUsers	Constant	Maximum supported concurrent users	≥ 200	Performance requirement
LocationUpdateInterval	Constant	Frequency of location updates	5 seconds	Performance requirement (F6)

Appendix B - Group Log

S. No.	Date	Timings	Venue	Description
1	08/01/2026	19:00 to 21:00	CCD	Several project ideas were evaluated, after which a unified campus food ordering and delivery system was proposed and finalized
2	11/01/2026	20:00 to 21:00	RM 3rd floor	Core objectives, target users, and high-level system features were identified, and the initial project registration form was completed and submitted
3	15/01/2026	18:00 to 19:30	RM 3rd floor	A detailed discussion was conducted on the Software Requirements Specification (SRS). Functional and non-functional requirements were identified, and the work was distributed among team members.
4	18/01/2026	16:00 to 16:30	DOAA Canteen	We met our mentor, Aditya Ojha, to discuss the project idea, the SRS document, and possible improvements.
5	23/01/2026	19:00 to 19:30	Online (Google Meet)	An online meeting was conducted to

				review the final SRS draft, consolidate feedback, and prepare the document for final submission.

- **Assignment of Work**
- **Ritik Ahirwar, Pravin Tanpure:** Prepared the initial SRS draft, including product scope, intended audience, definitions, document conventions, and references.
- **Vikas Saini, Abhimanyu Singh Rawat:** Developed the Overall Description section, covering product overview, system functionality, constraints, and assumptions.
- **Banoth Venkatamaniram, Kajal:** Defined external interface requirements, functional requirements, and use case models for core system operations.
- **Ashish Kumar Soni, Banoth Ramcharan:** Specified non-functional requirements, including performance, security, safety, and software quality attributes.
- **Aarsh Kaushik:** Integrated all contributions, added database and policy-related requirements, internationalization, reusability, and backup and recovery requirements, prepared the data dictionary, and finalized the complete SRS (Version 1.0)